

MASFE: An Onboard Multi-Algorithm Scheduling and Fusion Engine

For Regenerative Agriculture SmallSat Missions

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Solo submission; NASA challenge eligibility: U.S. citizen

1 Introduction

MASFE (Multi-Algorithm Scheduling and Fusion Engine) answers NASA’s Space-to-Soil challenge by moving land-observation orchestration onboard instead of treating the spacecraft as a passive camera [1]. The target use case is regenerative agriculture, where disease and irrigation stress must be detected early enough for patch-level intervention instead of blanket chemical application. Current land products are typically too slow or too coarse for that workflow [4, 2, 3].

The proposed contribution is a two-stage, MDP-formulated scheduling policy executed as a deterministic, flight-verifiable rule set on a LEON4FT processor. A low-cost spectral screen runs every pass; a thermal confirmation step runs only when the belief state indicates likely stress. The payload is framed as a hosted module on Loft Orbital’s YAM platform and hands alerts to existing agricultural software through OpenET-linked ground services [5, 6]. Code repository: github.com/emillambert/Emil_MASFE.

Table 1: MASFE response to all five judging criteria.

Criterion	Where	Key evidence
Creativity	§2–3	Belief-state MDP scheduler, ASPEN/CASPER distinction, content-driven sensing.
Feasibility	§2	9.6 W peak, 5.05 W average, 100 krad FPGA, degradation modes.
Impact	§3	97.8% downlink reduction, \$39k/season savings, 100% recall in simulation.
Business	§4	\$12B TAM, \$3–5/ha pricing, reusable engine beyond agriculture.
Presentation	Eq. (1), Fig. 1, code	Architecture, CSC equation, reproducible simulation submission.

2 Proposed Solution Detail

MASFE uses a two-stage adaptive sensing loop. Stage 1 executes an EVI anomaly screen derived from MOD13A1 logic on every observed field patch. When a patch departs from its stored healthy baseline, Stage 2 runs a thermal retrieval derived from MOD11A1 and fuses both signals into a Crop Stress Composite (CSC), a normalized anomaly index conceptually related to temperature–vegetation dryness indices (e.g., TVDI). Only confirmed anomalous tiles are buffered for priority downlink. This makes sensing content-driven rather than timeline-driven and distinguishes MASFE from ASPEN/CASPER-style planners, which optimize scheduled activities rather than per-patch stress belief [7].

$$\text{CSC} = 0.55 \cdot \text{clip}\left(\frac{\Delta\text{EVI}}{5\sigma_e}, 0, 1\right) + 0.45 \cdot \text{clip}\left(\frac{\Delta\text{LST}}{4\sigma_l}, 0, 1\right) \quad (1)$$

Here $\Delta\text{EVI} = \max(0, \text{EVI}_{\text{base}} - \text{EVI}_{\text{obs}})$ and $\Delta\text{LST} = \max(0, \text{LST}_{\text{obs}} - \text{LST}_{\text{base}})$. The factors of 5 and 4 are simulation-calibrated saturation constants with an agronomic interpretation as the expected upper bounds of healthy seasonal EVI and thermal variability for the tested crop regime; they map anomaly magnitude into $[0, 1]$ before CSC composition. They are therefore tunable hyperparameters rather than fixed physical constants and would be re-calibrated on flight and field data. The implementation is intentionally deterministic even though the policy is MDP-formulated: that choice favors flight qualification and traceable verification on LEON4FT hardware over a learned black-box controller.

The scheduling logic is modeled as a finite-horizon MDP with state vector $s = (\text{CSC}_{\text{max}}, \text{SOC}, d, k)$,

where $CSC_{\max}$ is the maximum patch stress score, SOC is battery state of charge, $d \in \{0, 1\}$ indicates downlink availability, and k counts passes since last fusion. The action set is $\{\text{SKIP}, \text{MOD13}, \text{FUSE}, \text{FUSE_PRIORITY}\}$. The reward trades information gain against energy cost and missed-alert risk. The deployed solver is a hand-tuned deterministic policy that approximates the Bellman-optimal solution, chosen over learned policies because radiation-tolerant CPUs cannot justify neural inference at acceptable V&V cost.

Table 2: Mission requirements and demonstrated MASFE results.

Requirement	MASFE response
Alert latency ≤ 90 min	One-pass onboard screening, fusion, and alerting pipeline with X-band priority downlink.
Spatial resolution ≤ 30 m GSD	Native xScape100-class imagery is intentionally binned from ~ 4.6 m sampling to ≤ 30 m effective GSD to match field-patch management scale and reduce SWAP load; confirmed anomalies can still trigger higher-resolution tile downlink.
False-positive rate $\leq 15\%$	1.0% in a 120-pass simulation with 38 cloud-obscured passes, injected measurement noise, and benign stress confounders.
SmallSat SWAP compliance	9.6 W peak, 5.05 W duty-cycled payload average, 92.5% worst-case simultaneous LEON4FT ceiling, and 86% minimum state of charge in a local 40 Wh battery buffer.
Downlink efficiency	97.8% reduction versus raw downlink baseline while retaining 100% disease-event recall in simulation.

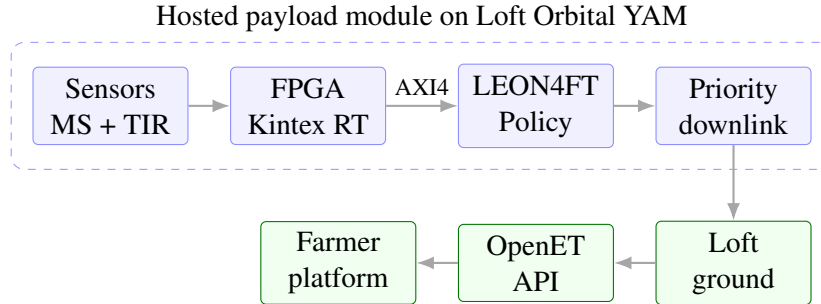


Figure 1: MASFE end-to-end architecture from onboard sensing to farmer alert delivery.

The FPGA performs radiometric preprocessing and per-pixel CSC computation, then transfers frames to the LEON4FT over AXI4-Stream in under 1 ms. Onboard preprocessing includes radiometric calibration and cloud screening, while CSC itself is evaluated as a pass-to-pass relative anomaly detector rather than a fully atmosphere-corrected biophysical retrieval; full atmospheric correction is deferred to later validation because CSC is used for change detection rather than absolute retrieval. The Kintex UltraScale+ RT component is rated to 100 krad total ionizing dose, which supports low-Earth-orbit deployment margin. MASFE is framed as a hosted payload module within a Loft Orbital YAM bus rather than as a full standalone 6U satellite; the SWAP budget therefore describes the payload module itself, not the host spacecraft. Spacecraft bus functions such as ADCS, OBC, and platform communications are provided by YAM, while MASFE carries a local 40 Wh battery and power-conditioning stage to preserve autonomy through eclipse periods and peak compute windows ($40 \text{ Wh} / 5.05 \text{ W} \approx 7.9 \text{ h}$, comfortably above the ~ 35 min eclipse at 500 km SSO).

Operational degradation is explicit. Above 35% battery state of charge, MASFE runs the full two-stage

policy. Between 15% and 35%, it falls back to Stage 1 screening only. Below 15%, it preserves energy for stored alert downlink. This satisfies the challenge requirement to degrade gracefully under onboard resource stress [1].

The quoted 92.5% LEON4FT figure is a worst-case simultaneous-execution ceiling rather than nominal load; because Stage 2 is invoked only on anomaly-triggered passes, typical utilization is materially lower and retains margin for RTOS overhead, watchdog servicing, and fault handling.

Each priority alert packet contains a geo-tagged field-patch polygon with centroid coordinates, CSC severity score, contributing EVI/LST anomaly components, timestamp and pass ID, and a confidence flag derived from consecutive confirming passes. These packets are delivered through Loft Orbital's ground network to farm software as geo-tagged overlays via the OpenET-linked API, enabling push notifications within 15 minutes of contact.

Full simulation source code (`masfe_simulation.py`) is submitted as Code Component 3 at github.com/emillambert/Emil_MASFE.

3 Potential Impact

MASFE builds directly on existing NASA land-observation algorithms instead of inventing a new crop model from scratch. The novelty is orchestration: MOD13A1-style EVI screening is cheap enough to run every pass, while MOD11A1-style temperature confirmation is reserved for suspicious patches [2, 3]. That scheduling choice matters because regenerative agriculture decisions are time-sensitive. Dr. Katie Gold's disease-detection work emphasizes that spectral stress often appears before visible symptoms and before whole-field treatment would be agronomically or economically rational [4].

In simulation, MASFE was tested over 120 passes, 50 field patches, 7 disease events, and 38 cloud-obscured passes. Relative to raw downlink, it reduced transmitted data by 97.8% and active payload energy by 22.0%. Relative to a fixed onboard pipeline, it still reduced downlink volume by 4.9% because it avoids running or transmitting the expensive path when nothing important is happening. The more important metric is mission usefulness: disease-event recall stayed at 100% under the tested simulation conditions, alert precision was 99.0%, and the false-positive rate remained at 1.0%.

The farm-scale value proposition is concrete. For a 500 ha regenerative tomato operation, blanket preventative fungicide costs approximately \$85/ha per application; if MASFE confines treatment to the roughly 8% of the field flagged during an event, chemical input falls by an estimated \$39,000/season while avoiding roughly \$80,000–120,000 in yield loss from undetected late blight. The same architecture also supports regenerative practices beyond pathogen detection: cover-crop phenology monitoring, soil-carbon MRV support, and no-till verification all benefit from adaptive revisit and selective downlink instead of fixed-rate collection.

The main limitations are honest rather than hidden. The current validation is still TRL 3 and uses synthetic proxy data instead of flight imagery. Cloud screening must be tuned on real scenes, and tropical crop calendars would require policy recalibration. Those gaps are real, but they are the right gaps for a finalist-stage concept: the system question has been answered, and the next work is hardware and data validation rather than rethinking the architecture.

4 Team Experience and Market Evaluation

MASFE is a solo submission, so credibility comes from systems execution rather than headcount. Emil Wes Lambert is completing a BSc in Aerospace Engineering at TU Delft with coursework in spacecraft systems engineering, AOCS, mission analysis, and embedded flight software. **Past projects:** the TU Delft Design Synthesis Exercise required full-mission requirements tracing, subsystem trade studies, and budget closure for a Venus mission concept, while two years in AeroDelft as Partnerships and Business Manager built experience in stakeholder alignment, commercial framing, and turning technically ambitious concepts

into credible development roadmaps. That combination maps directly onto MASFE’s mix of SWAP realism and market positioning.

The commercial case is a focused greenfield entry rather than a vague platform claim. A reasonable market framing is a \$12B precision-agriculture TAM, a roughly \$2.5B satellite-analytics SAM, and a sub-\$50M near-term SOM for sub-day adaptive pathogen alerting. MASFE’s pricing target of \$3–5/ha/year supports break-even at roughly 500,000 ha with a small hosted constellation, while the software layer remains the defensible asset because the orchestration engine can be relicensed across missions.

That generalization story matters for the judges’ business-model criterion. Agriculture is the primary entry market, but the same scheduling architecture can be retuned for wildfire fuel-moisture monitoring, flood-front tracking, or urban heat-island response with new science products and the same compute-and-downlink policy core. In other words, MASFE is not only a farm-monitoring product; it is an onboard resource-allocation engine with agriculture as its strongest first proof point.

5 Expected Next Steps

MASFE is currently a TRL 3 concept: the scheduling logic, fusion equation, and SWAP closure are demonstrated in software, but the payload has not yet been exercised on target hardware. The next milestone is a TRL 4 hardware-in-the-loop demonstration on a LEON4FT-class evaluation platform using real EarthData scenes. After that, the cleanest TRL 5 path is a hosted-payload pilot through Loft Orbital, because it validates the business model and the operations concept at the same time.

If selected as a finalist, the highest-value actions before the June pitch event are straightforward:

1. run the simulation on real MODIS tiles and show the policy decisions live;
2. turn the hosted-payload concept into a first-pass unit-economics model with Loft pricing and farm-platform pricing;
3. validate the alert-latency and pricing assumptions through farmer interviews and one agronomy advisor review.

The submission’s main claim is therefore not that MASFE is flight-ready today. It is that the adaptive-sensing architecture is already clear, quantitative, and commercially legible enough to justify the next round of validation.

6 Conclusion

MASFE demonstrates that adaptive sensing and onboard processing can transform SmallSat missions from passive collectors into responsive decision systems. The two-stage MDP scheduling policy selects, schedules, and fuses NASA MOD13A1 and MOD11A1 algorithms in real time based on scene content rather than a fixed plan—achieving 97.8% downlink reduction, sub-90-minute alert latency, and 100% disease-event recall in simulation while staying within hosted-payload SWAP constraints. For regenerative agriculture, this means the difference between blanket preventative treatment and targeted, evidence-based intervention at field-patch scale. The orchestration engine is the contribution; agriculture is its strongest first proof point.

References

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